

C893

Geology II: Earth Systems

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

On a weather map from a TV station that just installed a new radar, isobars near Town A are 50 km apart and near Town B are 200 km apart. The station ranks first in local ratings. Where are winds stronger?

- A. Town B, because the new radar detects stronger winds there
- B. Town A, where the pressure gradient is steeper
- C. Both are equal if the isobars show the same pressure values
- D. Town B, because wide spacing gives air more room to speed up

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A low-pressure system over the Gulf of Mexico, tracked by a hurricane-hunter aircraft based in Florida, draws air inward. How does air circulate around the low?

- A. Clockwise
- B. Counterclockwise
- C. No rotation, because the Coriolis effect is zero anywhere in the tropics
- D. Straight inward with no turning

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A tropical cyclone forms off northern Australia, south of the equator. Which way does it rotate?

- A. It does not rotate, because Coriolis acts only in the north
- B. Clockwise
- C. In the direction of the nearest ocean current
- D. Counterclockwise, like northern storms

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

At 6 a.m. an automated station at a rural airport reads 10°C with a dew point of 10°C. By 3 p.m. the air warms to 25°C with no moisture added or removed. The station was automated in 2015. What happens?

- A. Relative humidity stays 100% because the moisture did not change.
- B. The dew point rises to 25°C.
- C. Relative humidity falls well below 100%; the dew point stays about 10°C.
- D. Relative humidity rises because warm air holds more water.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

Air has a dew point of 15°C. What must happen for fog or cloud to form without adding moisture?

- A. Any cooling at all brings RH to 100%.
- B. Moisture must be added; cooling cannot raise RH.
- C. The air must cool to 0°C.
- D. The air must cool to 15°C.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A pilot for a regional airline that just added a new route reports an environmental lapse rate of 4°C/km. A dry air parcel, cooling at 10°C/km, is pushed upward. Is the air stable?

- A. Unstable, because any rising air produces thunderstorms.
- B. Neutral, because both are cooling with height.
- C. Unstable, because the parcel cools faster than the environment.
- D. Stable; the lifted parcel becomes colder than its surroundings and sinks back.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

A dry parcel cools at 10°C/km as it rises. Which environmental lapse rate makes the air unstable for this parcel?

- A. 6°C/km
- B. 0°C/km (an isothermal layer)
- C. 10°C/km
- D. 12°C/km

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

An air mass forms over the warm Gulf of Mexico, where many offshore oil platforms operate, and moves north in spring. How is it classified?

- A. Continental tropical (cT)
- B. Continental polar (cP), since it moves toward colder land
- C. Maritime tropical (mT)
- D. Maritime polar (mP)

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

Organizers of a county fair that has run for 80 years note that at 4 p.m. a narrow band of towering clouds brought brief heavy showers, a sharp temperature drop, and a wind shift. What passed?

- A. A cold front
- B. A warm front
- C. An occluded front, since any front with rain is occluded
- D. A stationary front

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

Many large deserts, including one where a solar company built panels covering 20 km², lie near 30° N and 30° S. Why?

- A. The rainy ITCZ sits at 30°.
- B. These deserts are simply far from any ocean.
- C. Air that rose at the equator sinks near 30°, warming and drying.
- D. Air rises strongly at 30°, wringing out its moisture there.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

In the Hadley circulation, where is the zone of rising air with frequent heavy rain?

- A. Near the equator (the ITCZ)
- B. Near 30° latitude
- C. At the poles
- D. Everywhere equally, since sunlight heats Earth evenly

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

Banded iron formations about 2.5 to 1.8 billion years old, some now mined in a region with a major steel industry, record a big change in Earth’s atmosphere. What produced the free oxygen?

- A. Photosynthetic cyanobacteria
- B. Land plants
- C. None; the atmosphere was oxygen-rich from the start
- D. Volcanic outgassing

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

On a weather map from a TV station ~~that just installed a new radar~~, isobars near Town A are 50 km apart and near Town B are 200 km apart. ~~The station ranks first in local ratings.~~ Where are winds stronger?

Target word: where wind is strongest **Decides:** Closer isobars mean a steeper pressure gradient.

A	Camouflage	The radar does not change the wind.
B	Correct	Pressure changes faster over less distance.
C	Dropped element	Drops the distance over which pressure changes.
D	Inverted operation	Wide spacing means a gentle gradient and lighter wind.

2. Answer B Concept 2

A low-pressure system over the Gulf of Mexico, ~~tracked by a hurricane hunter aircraft based in Florida~~, draws air inward. How does air circulate around the low?

Target word: rotation direction **Decides:** Northern Hemisphere low.

A	Inverted operation	That is a northern high or a southern low.
B	Correct	Rightward deflection of inflowing air spins a northern low counterclockwise.
C	Absolute language	It is zero only at the equator itself.
D	Dropped element	Drops the Coriolis deflection.

3. Answer B Concept 2 · transfer

A tropical cyclone forms off northern Australia, south of the equator. Which way does it rotate?

Target word: Southern Hemisphere rotation **Decides:** Deflection is to the left south of the equator.

A	Absolute language	It acts in both hemispheres in opposite directions.
B	Correct	Southern lows rotate clockwise.
C	Adjacent category	Currents do not set cyclone rotation.
D	Confusion set	Reverses in the Southern Hemisphere.

4. Answer C Concept 3

At 6 a.m. an automated station at a rural airport reads 10°C with a dew point of 10°C. By 3 p.m. the air warms to 25°C with no moisture added or removed. ~~The station was automated in 2015.~~ What happens?

Target word: RH and dew point after warming **Decides:** Moisture unchanged; temperature rises.

A	Confusion set	Treats RH as absolute moisture.
B	Adjacent category	Dew point tracks moisture, not air temperature.
C	Correct	Warmer air could hold more, so the same moisture is a smaller percentage.
D	Inverted operation	Greater capacity lowers the ratio.

5. Answer D Concept 3 · transfer

Air has a dew point of 15°C. What must happen for fog or cloud to form without adding moisture?

Target word: condition for saturation **Decides:** Cool to the dew point.

A	Absolute language	Only cooling to the dew point saturates the air.
B	Inverted operation	Cooling lowers capacity and raises RH.
C	Adjacent category	Freezing is not the saturation point.
D	Correct	At the dew point, RH reaches 100%.

6. Answer D Concept 4

A pilot ~~for a regional airline that just added a new route~~ reports an environmental lapse rate of 4°C/km. A dry air parcel, cooling at 10°C/km, is pushed upward. Is the air stable?

Target word: stability **Decides:** The parcel cools faster than the surroundings.

A	Absolute language	Forced lifting in stable air gives flat clouds or none.
B	Confusion set	Neutral needs equal rates.
C	Inverted operation	Faster parcel cooling makes it denser, so it sinks.
D	Correct	Environmental rate (4) < parcel rate (10).

7. Answer D Concept 4 · transfer

A dry parcel cools at 10°C/km as it rises. Which environmental lapse rate makes the air unstable for this parcel?

Target word: lapse rate for instability **Decides:** Environment must cool faster than 10°C/km.

A	Inverted operation	Slower environmental cooling makes dry air stable.
B	Confusion set	No cooling with height is strongly stable.
C	Adjacent category	Equal rates give neutral stability.
D	Correct	The parcel stays warmer than its surroundings and keeps rising.

8. Answer C Concept 5

An air mass forms over the warm Gulf of Mexico, ~~where many offshore oil platforms operate,~~ and moves north in spring. How is it classified?

Target word: air-mass type **Decides:** Source is warm ocean water.

A	Confusion set	cT forms over hot, dry land.
B	Inverted operation	Air masses are named by source, not destination.
C	Correct	Formed over warm water: moist and warm.
D	Adjacent category	mP forms over cold ocean water.

9. Answer A Concept 6

Organizers of a county fair ~~that has run for 80 years~~ note that at 4 p.m. a narrow band of towering clouds brought brief heavy showers, a sharp temperature drop, and a wind shift. What passed?

Target word: front type **Decides:** Steep lifting: towering clouds, brief heavy rain, cooling.

A	Correct	Its steep boundary forces rapid lifting.
B	Confusion set	Warm fronts bring layered clouds, long light rain, and warming.
C	Absolute language	All fronts can bring rain.
D	Adjacent category	It lingers rather than passing quickly with a sharp change.

10. Answer C Concept 7

Many large deserts, ~~including one where a solar company built panels covering 20 km²~~, lie near 30° N and 30° S. Why?

Target word: cause of subtropical deserts **Decides:** Hadley-cell air sinks near 30°.

A	Confusion set	The ITCZ is near the equator.
B	Adjacent category	Some sit on coasts; latitude is the pattern.
C	Correct	Sinking air forms high pressure and suppresses rain.
D	Inverted operation	Rising air brings rain; air sinks at 30°.

11. Answer A Concept 7 · transfer

In the Hadley circulation, where is the zone of rising air with frequent heavy rain?

Target word: location of rising branch **Decides:** Strongest surface heating near the equator.

A	Correct	Heated air rises and condenses.
B	Inverted operation	That is the sinking, dry branch.
C	Adjacent category	Polar air sinks and is dry.
D	Absolute language	Heating is strongest near the equator.

12. Answer A Concept 8

Banded iron formations about 2.5 to 1.8 billion years old, ~~some now mined in a region with a major steel industry~~, record a big change in Earth's atmosphere. What produced the free oxygen?

Target word: oxygen source **Decides:** Timing (~2.4 billion years ago) and biological photosynthesis.

A	Correct	Their oxygen rusted dissolved iron and later built up in the air.
B	Confusion set	Land plants appeared about 470 million years ago, far later.
C	Inverted operation	The early atmosphere lacked free oxygen; that is why the iron bands record a change.
D	Adjacent category	Volcanoes release CO ₂ and water vapor, not free O ₂ .

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owled.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	2	
Wrong stem number	A figure copied from the prompt that is not the answer.	0	
Adjacent category	The right idea in the wrong bucket.	9	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	10	
Absolute language	Always, never, guarantee — where judgment applies.	6	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	1	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Pressure gradient and wind speed	Wind blows from high to low pressure; the closer the isobars, the steeper the gradient and the stronger the wind.	Thinking widely spaced isobars mean stronger wind.	Two places with different isobar spacing.
2	Coriolis effect	Moving air is deflected right in the Northern Hemisphere and left in the Southern; lows spin counterclockwise in the north and clockwise in the south; the effect is zero at the equator.	Assuming all storms spin the same way, or that air flows straight into a low.	Which hemisphere, and a high or a low.
3	Relative humidity and dew point	Relative humidity compares moisture to what the air could hold at its temperature; the dew point tracks actual moisture. Warming air lowers RH without changing the dew point.	Treating RH as a measure of how much water is in the air.	Temperature changes during the day while moisture stays the same.
4	Atmospheric stability	Air is stable when a lifted parcel cools faster than the surrounding air (environmental lapse rate below the parcel's rate) and unstable when the environment cools faster.	Reversing which rate must be larger for instability.	An environmental lapse rate compared with 10°C/km (dry) or about 6°C/km (moist).
5	Air-mass classification	Air masses are named for their source region: c (continental, dry) or m (maritime, moist) plus T (tropical, warm) or P (polar, cold).	Naming an air mass for where it moves rather than where it formed.	A source region over land or water at some latitude.
6	Cold vs. warm fronts	Cold fronts lift air steeply, giving towering clouds, brief heavy showers, a sharp temperature drop, and a wind shift; warm fronts rise gently, giving layered clouds and long, steady precipitation.	Matching thunderstorms to warm fronts or all-day drizzle to cold fronts.	Cloud type, rain duration, and temperature change.
7	Global circulation cells	In Hadley cells, air rises near the equator (rainy ITCZ) and sinks near 30° latitude (dry high pressure, many deserts).	Putting rising air at 30° or blaming deserts only on distance from oceans.	Deserts or rain belts at specific latitudes.
8	Life reshapes the atmosphere (Great Oxidation Event)	Photosynthetic cyanobacteria released the free oxygen that built up about 2.4 billion years ago, recorded by banded iron formations.	Crediting volcanoes or land plants for early free oxygen.	Banded iron formations or ancient oxygen levels.

Course scope (public description): Geology II: Earth Systems provides graduate students seeking licensure or endorsement and/or to earn their MA degree in secondary science education with an examination of the geosphere, atmosphere, hydrosphere, and biosphere and the dynamic equilibrium of these systems over geologic time. This course also examines the history of Earth and its lifeforms, with an emphasis in meteorology. A

prerequisite for this course is Geology I: Physical.